VIT

Vellore Institute of Technology

CHENNAI

Continuous Assessment Test (CAT) – II OCTOBER 2025

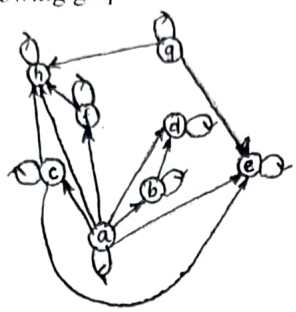
Programme	:	B.Tech	Semester	:	FALL SEMESTER 2025 – 26
Course Code & Course Title	:	BMAT205L & Discrete Mathematics and Graph Theory	Slot	:	D1+TD1+TDD1
Faculty	:	Dr. Radha S., Dr. Devi Yamini S., Dr. Poulomi De, Dr. Somnath Bera, Dr. Sankarsan Tarai, Dr. Basua Debananda, Dr. Gayathri M., Dr. Mohd Imran Idrisi, Dr. Sriraman R., Dr. Rajesh Kumar Mohapatra, Dr. Pavithra R., Dr. Amit Kumar Rahul	Class Number	:	CH2025260100755, CH2025260100757, CH2025260100759, CH2025260100760, CH2025260100761, CH2025260100762, CH2025260100763, CH2025260100764, CH2025260100765, CH2025260100766, CH2025260100768, CH2025260100769
Duration	:	90 Minutes	Max. Mark	:	50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1.		Use the method of generating function to solve the recurrence relation $p_n = 2p_{n-1} + p_{n-2}; n \geq 2; p_0 = 0, p_1 = 1$. (Pell's Number)	10	CO3	K3
2.	(a)	Express the Boolean function $f(P,Q,R) = PQ + PR + PQR$ in both canonical Sum of Products (SOP) form and canonical Product of Sums (POS) form.	4	CO4	K3
	(b)	Given a partial ordered set, $S = \{a,b,c,d,e,f,g,h\}$ with the relation R, whose matrix representation after removal of reflexive and transitive edges is $R_s = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{array} \begin{array}{c} \begin{array}{cccccccc} a & b & c & d & e & f & g & h \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$ Determine the given POSET is a lattice. If it is, check whether it is complemented.	6		

3.	Consider the following graph  (i) Write the relation R from the graph and prove that it is POSET. **(3 Marks)** (ii) Draw the Hasse diagram for the given directed graph. **(2 Marks)** (iii) Find the maximal, minimal, greatest and least element of the graph (if it exists) **(2 Marks)** (iv) Find the upper bound, lower bound, LUB and GLB for the subsets $\{b, f\}$ and $\{c, e, f\}$. **(3 Marks)**	10	CO4	K4
4.	To improve the connectivity for five cities: Chennai, Delhi, Mumbai, Kolkata and Bengaluru, an airline decides to operate direct flights between every pair of cities. (a) Model this situation as a graph, G. **(2 Marks)** (b) Check whether the graph, G is regular and justify your answer. **(2 Marks)** (c) List any three cyclic subgraphs of the graph, G **(3 Marks)** (d) Identify any one spanning subgraph of the graph, G. **(1 Mark)** (e) Give an example of a walk of length 4 in the graph, G. **(1 Mark)** (f) Give an example of a trail of length 5 in the graph, G. **(1 Mark)**	10	CO5	K4
5.	(a) A social media platform is analysing user interactions through post tags. Four users Arjun, Priya, Rohan and Kavya tag each other in posts as follows: Arjun tagged Priya and Rohan; Priya tagged Rohan; Rohan tagged Kavya; Kavya tagged Arjun. This has modelled as Graph, G_1. Determine whether the constructed graph G_1 is isomorphic to the graph G_2 whose incident matrix is $I_{G_2} = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$ Where v_1, v_2, v_3, v_4 represents vertices of G_2 and e_1, e_2, e_3, e_4, e_5 represents edges of G_2.	6	CO5	K4
	(b) A smart city traffic network connects four intersections A, B, C, D with the adjacency matrix $Adj_Matrix = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$. (i) Draw the graph corresponding to the adjacency matrix. **(2 Marks)** (ii) Determine whether the graph is Eulerian graph or/and Hamiltonian graph. Justify. **(2 Marks)**	4		

**********All the best **********